

FJFI - Department of Physics
02MECHZ - Mechanics
Exam

31.01.2025
13:00pm - 16:00pm

Name	SOLUTIONS				
Question	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 180 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
4. Calculators and use of any class materials are not allowed.
5. Pencils and erasable pens are not allowed.
6. Write your answers legibly. Illegible answers will not be scored.
7. The grading of your exams will be based on the answers you have written on the five stapled sheets of paper you are holding in your hand.

Question 1

A particle of mass m moves under the influence of a central force $\vec{F}(r) = (-kr + \frac{C}{r^3}) \hat{r}$, where $k > 0$ and $C > 0$ are constants, r is the radial distance from the origin, and $\hat{r}$ is the unit vector in the radial direction.

- Explain in very simple terms (preferably by using the definition of torque) why the angular momentum of the particle is conserved. — 0.4 points
- Find the effective potential of the particle is given as $V(r) = U(r) + \frac{l^2}{2mr^2}$, where $U(r)$ is the potential of the particle and l is the angular momentum. — 0.4 points
- Find the equilibrium point(s) of the particle. What values can C take for an equilibrium point to exist? — 0.2 points

Solution:

a) Position vector in spherical coordinates: $\vec{r} = r \hat{r}$ where $\hat{r}$ is the unit vector in radial direction

$$\vec{\tau} = \vec{r} \times \vec{F} = (r \hat{r}) \times (-kr - \frac{C}{r^3}) \hat{r} = r(-kr - \frac{C}{r^3}) \hat{r} \times \hat{r} = 0$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L} : \text{constant.}$$

b) $U(r) = - \int F dr = - \int (-kr + \frac{C}{r^3}) dr = \int kr dr - C \int r^{-3} dr = \frac{1}{2} k r^2 + \frac{1}{2} C \frac{1}{r^2}$

$$V(r) = \frac{1}{2} k r^2 + \frac{1}{2} C \frac{1}{r^2} + \frac{l^2}{2mr^2} = \frac{1}{2} k r^2 + \frac{1}{2} \left(C + \frac{l^2}{m} \right) \frac{1}{r^2}$$

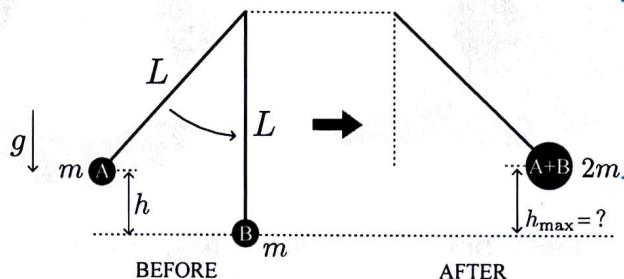
c) $\frac{dV}{dr} = 0 \Rightarrow kr - \frac{2}{2} \left(C + \frac{l^2}{m} \right) \frac{1}{r^3} = 0 \Rightarrow kr^4 = C + \frac{l^2}{m} \Rightarrow r = \left(\frac{C + \frac{l^2}{m}}{k} \right)^{1/4}$

$$k > 0, C > 0, l^2 > 0, m > 0 \Rightarrow r \text{ exists if } C + \frac{l^2}{m} \geq 0 \Rightarrow C \geq -\frac{l^2}{m}$$

! A common mistake: In central-force motion problem $F(r)$ becomes zero at equilibrium point only if the trajectory is circular. In general, you cannot use $F(r) = 0$ to find equilibrium points in central-force motion problem.

Question 2

Consider two identical pendulums as shown in the figure, each consisting of a mass m attached to a string of length L suspended from the same point. The pendulums are initially at rest. The pendulum A is released from a height h relative to B, while the pendulum B is initially stationary.



- Determine the velocity of the pendulum A just before it collides with the pendulum B. - 0.4 points
- After the collision, the two pendulums stick together and swing as one system. Calculate the maximum height reached by the combined system after the collision. - 0.6 points

Solution:

$$a) \quad mgh = \frac{1}{2}mv^2 \Rightarrow \boxed{v = \sqrt{2gh}}$$

b) "... stick together..." : inelastic collision $\Rightarrow$ Total mechanical energy is not conserved.

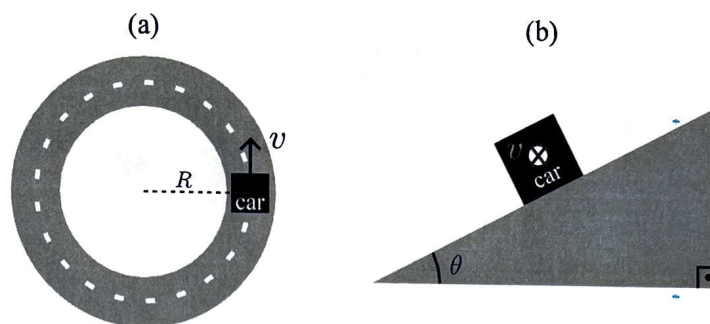
But linear momentum is conserved:

$$\Rightarrow p_i = p_f \Rightarrow mV + 0 = 2mV_f \Rightarrow V_f = V/2$$

After collision, and A and B stick together, we can use the conservation of total energy:

$$\frac{1}{2}(2m)V_f^2 = (2m)gh_{\max} \Rightarrow h_{\max} = \frac{V_f^2}{2g} = \frac{V^2}{8g} = \frac{2gh}{8g} = \boxed{\frac{h}{4}}$$

Question 3



A car of mass m is moving at a speed v around a curve with a radius R . The coefficient of static friction between the tires and the road is μ_s , and the gravitational acceleration is g .

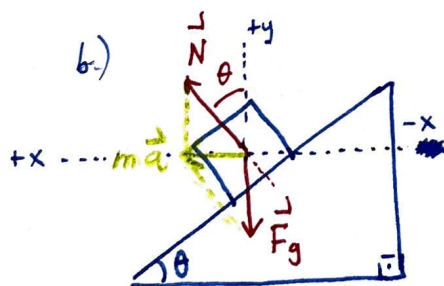
- Determine the maximum speed $v_{\max}$ at which the car can navigate the curve safely to avoid slippage as shown in Figure (a). — 0.7 points
- If the road is banked as shown in Figure (b) at an angle θ with respect to the horizontal, find the maximum speed for which the car can take the curved path safely without depending on friction. — 0.3 points

Note: All the other other external forces, such as air resistance, are negligible.

Solution:

a) Centripetal force = static friction = $f_s = \mu_s N = \mu_s mg$

Newton's 2nd law for circular motion: $f_s = ma \Rightarrow \mu_s mg = m \frac{v_{\max}^2}{R} \Rightarrow \boxed{v_{\max} = \sqrt{\mu_s g R}}$



$\vec{F}_g$ and $\vec{N}$ are the only forces acting on the car. According to the question the car does a circular motion with radius R and velocity V .

Therefore $\vec{F}_g$ and $\vec{N}$ have to add up to the centripetal force only. $\therefore \vec{F}_g + \vec{N} = \text{centripetal} = m\vec{a}$

★ The vectorial equation $\vec{F}_g + \vec{N} = m\vec{a}$ can be written in terms of its components for an arbitrarily chosen coordinate axes. A useful choice, which would allow us to solve the problem easily is shown in the figure. According to that:

In y-direction:

$$\vec{F}_g + \vec{N} = m\vec{a} \quad (\vec{a} \text{ has no component on y and})$$

$$-mg + N \cos \theta = 0$$

In x-direction

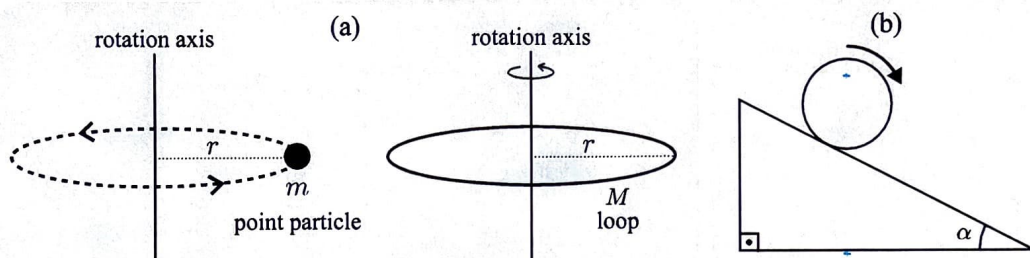
$$\vec{F}_g + \vec{N} = m\vec{a}$$

$$0 + N \sin \theta = ma$$

$$\frac{N \sin \theta}{N \cos \theta} = \frac{ma}{mg} \Rightarrow g \tan \theta = a = \frac{v^2}{R} \Rightarrow \boxed{v = \sqrt{Rg \tan \theta}}$$

central acceleration

Question 4

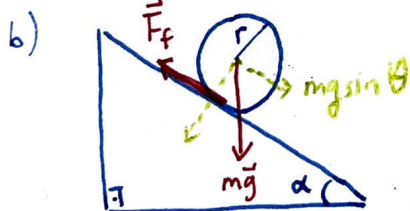


- (a) Given that the rotational inertia of a point particle as shown in the Figure (a) is mr^2 , explain why the rotational inertia of a uniform loop would be Mr^2 ? — 0.2 points
- (b) Calculate the acceleration of a uniform loop as it rolls without sliding on a rough inclined plane inclined at an angle α to the horizontal as shown in Figure (b), in a uniform gravitational field of magnitude g . — 0.8 points

Solution:

- a) Divide the loop into smaller parts with mass Δm (so that $\sum \Delta m = M$). Each part's contribution to rotational inertia will be the same $\rightarrow \Delta m r^2$ which is the same as that of a point particle. Therefore, the total rotational inertia will be

$$I = \sum \Delta m r^2 = r^2 \sum \Delta m = \boxed{Mr^2}$$



Newton's 2nd law written for the motion of COM:

$$mg \sin \theta - F_f = m a_{\text{com}} \quad (\text{we neglected the rotational motion in here})$$

Newton's 2nd law in the rotational form:

$$\tau = I \ddot{\theta}, \quad (\tau = F_f r, \quad y = r\theta, \quad \dot{y} = r\dot{\theta}, \quad \ddot{y} = r\ddot{\theta})$$

$$\Rightarrow r F_f = (mr^2) \ddot{\theta}$$

$$F_f = m \ddot{y} = m a_{\text{com}}$$

$$m(g \sin \theta - a_{\text{com}}) = m a_{\text{com}}$$

$$g \sin \theta = 2 a_{\text{com}} \Rightarrow \boxed{a_{\text{com}} = \frac{g \sin \theta}{2}}$$